Block practice · Python: subprograms, files and the 20-mark program

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Subprograms and strings: complete the function and its call

Read and write text files line by line

The final 20-mark program

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Practice workbook · Subprograms and strings: complete the function and its call

A short Paper 2 program built around subprograms and string handling: completing a function or procedure body, passing arguments, returning a value, calling it from the main program or a menu, and string work inside it (slicing, joining with numbers, reversing, checking length with re-input).

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-CODE-07

1. Question 4 · part c · 6 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

4 Programmers write code that keeps track of values, is efficient and uses subprograms.

(a) Global variables are different from local variables.

(i) State where a **global** variable is created.

(1)

(ii) State where a **local** variable is accessible.

(1)

(b) An algorithm reports the range that an integer falls within.

Integers from 0 to 39 are in the low range.

Integers from 40 to 69 are in the middle range.

Integers from 70 to 100 are in the high range.

Figure 4 and **Figure 5** show two algorithms that solve this problem.

```
1  SEND 'Enter an integer: ' TO DISPLAY
2  RECEIVE theNumber FROM (INTEGER) KEYBOARD
3  IF (theNumber < 40) THEN
4      SEND 'Low range' TO DISPLAY
5  ELSE
6      IF (theNumber < 70) THEN
7          SEND 'Middle range' TO DISPLAY
8      ELSE
9          SEND 'High range' TO DISPLAY
10     END IF
11 END IF
```

Figure 4

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8

P 7 5 7 3 7 A 0 8 1 6

1. Question 4 · part c · 6 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

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```
1   SEND 'Enter an integer: ' TO DISPLAY
2   RECEIVE theNumber FROM (INTEGER) KEYBOARD
3   IF (theNumber < 40) THEN
4       SEND 'Low range' TO DISPLAY
5   END IF
6   IF (theNumber > 39) AND (theNumber < 70) THEN
7       SEND 'Middle range' TO DISPLAY
8   END IF
9   IF (theNumber > 69) THEN
10      SEND 'High range' TO DISPLAY
11  END IF
```

Figure 5

Explain the reason that the algorithm in **Figure 4** is more efficient than the algorithm in **Figure 5**.

(2)

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1. Question 4 · part c · 6 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

(c) A program is required to create a new key.

The program takes two inputs.

The first input is a four-character string.

The second input is a whole number.

The key is constructed by joining the first two characters from the string, the number and the final two characters from the string.

When the user enters the four-character string **abcd** and the integer **123**, the program must construct and display the new key **ab123cd**

Open **Q04c** in the code editor.

Amend the code to:

- complete the subprogram to construct the new key
- complete the call to the subprogram.

Do not add any further functionality.

Save your code as **Q04cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 4 = 10 marks)

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1. Question 4 · part c · 6 marks · Working and answer

Working and answer

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2. Question 4 · part b · 7 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

4 Gianluca is a schoolteacher. He is in charge of a year group of pupils.

(a) He has started to create a program to analyse test results.

Open **Q04a** in the code editor.

Amend the code to complete the 'if statement' used to produce the outputs described in this table.

Condition	Output text
English test score was less than 40; Maths test score was less than 50	Student failed both tests
English test score was less than 40 or Maths test score was less than 50	Student failed one test
English test score was at least 80; Maths test score was at least 85	Student passed both tests with distinction
Other conditions	Student passed both tests

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(6)

(b) Gianluca has started to create a program to analyse pupil attendance.

He wants to use a menu system with subprograms.

Open **Q04b** in the code editor.

The program must include subprograms to:

- display the **names** of students whose percentage attendance was less than 75%
- count and display the **number** of students whose percentage attendance was 90% or higher.

Complete the program code to implement these requirements using the structure given in **Q04b**.

Do not add any further functionality.

Save your code as **Q04bFINISHED** with the correct file extension for the programming language.

(7)

(Total for Question 4 = 13 marks)

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2. Question 4 · part b · 7 marks · Working and answer

Working and answer

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3. Question 4 · part a · 7 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

4 Zak sells snacks at an after-school club.

(a) He wants a program that will hold details of the products he sells.

The program must:

- generate a five-character product code that:
  - starts with the first three letters of the product name
  - ends with a random number between 10 and 30
- display the product code followed by the product name.

Open **Q04a** in the code editor.

Write the program.

You must use the structure given in the file **Q04a** to complete the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(7)

(b) Zak plans to implement a binary search algorithm to search a table of products.

(i) Explain **one** advantage of a binary search compared to a linear search.

(2)

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3. Question 4 · part a · 7 marks · Working and answer

Working and answer

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4. Question 4 · part a · 11 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

4 Programs can be used to manipulate data.

(a) A program is needed to convert an 8-bit binary number into denary.

The program must:

- receive the input of a binary pattern and validate that it has exactly eight characters (there is no need to validate that it only contains 0s and 1s)
- request the input again if the binary pattern does not have exactly eight characters
- when a valid binary pattern is input:
  - convert the binary pattern into its denary equivalent
  - print a message that contains the binary pattern and its denary equivalent
- include a comment to explain how the conversion process works.

HINT

Denary placeholder	128	64	32	16	8	4	2	1
Binary number	1	0	1	1	0	1	0	0
Denary number	128	0	32	16	0	4	0	0

128 + 32 + 16 + 4 = 180

Open **Q04a** in the code editor.

Write a program to implement these requirements.

You must use the structure given in **Q04a** to write the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(11)

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10

P 6 9 2 9 8 A 0 1 0 1 6

4. Question 4 · part a · 11 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

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(b) Open **Q04b** in the code editor.

Describe the purpose of the program.

(2)

(Total for Question 4 = 13 marks)

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4. Question 4 · part a · 11 marks · Working and answer

Working and answer

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5. Question 4 · part b · 8 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

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4 Data is encoded for encryption and for identifying records in databases.

(a) A Caesar cipher is one method of encryption.

(i) Complete the table to show the result of applying a Caesar cipher.

(2)

Plaintext	Shift	Ciphertext
PIXEL	−4	
CLOUD		FORXG

(ii) Plaintext is encrypted using a Caesar cipher.

Compare the ciphertexts produced for the plaintext **GOLD** by:

- a shift of −6 followed by a shift of +8
- a single shift of +2.

(2)

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(iii) Ann has written a Caesar cipher algorithm.

When the algorithm encrypts the plaintext **BYTE** with a shift of +5 the result is **GVYJ**.

This is incorrect.

Explain the error.

(2)

5. Question 4 · part b · 8 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

(b) A program is needed to create a key for a database.

A user enters a two-letter word, a whole number and a decimal number.

The program must ensure the word is only two characters long.

The program must display an error message when the word is not two characters long.

A key is generated from the whole number, the reversed word and the whole number part of the decimal number.

Figure 4 shows the input values and a valid key.

```
Enter a word: at
Enter a whole number: 12
Enter a decimal number: 7.89
12ta7
```

Figure 4

Open **Q04b** in the code editor.

Write the program.

Do not add any further functionality.

Save your code as **Q04bFINISHED** with the correct file extension for the programming language.

(8)

(Total for Question 4 = 14 marks)

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5. Question 4 · part b · 8 marks · Working and answer

Working and answer

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6. Question 2 · part c.ii · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part c.ii only.

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2 Programs can process numerical data.

(a) Identify the method used to check input data matches set rules.

(1)

- ☒ **A** Abstraction
- ☒ **B** Encryption
- ☒ **C** Iteration
- ☒ **D** Validation

(b) **Figure 2** shows a section from an employee payslip produced by a computer program.

Pay Rate	Hours		Total Pay
£ 18.75	19		£ 356.25
	Tax	20%	£ 71.25
		Income	£ 285.00

Figure 2

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(i) Complete the table to give **one** data type shown in **Figure 2** and give an example value.

(2)

Data type	
Example	

(ii) Complete the table to give an input, a process and an output used in the program to produce the payslip.

(3)

Input	
Process	
Output	

6. Question 2 · part c.ii · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part c.ii only.

(c) Numerical values can be represented in denary, binary and hexadecimal.

A program converts denary values in the range 0 to 255 into hexadecimal.

(i) The program is tested using appropriate integer values.

Complete the table to show an example of normal, boundary and erroneous test data for this program.

(2)

	Test data
Normal	
Boundary	
Erroneous	

(ii) The program uses subprograms to do the conversion.

Open **Q02cii** in the code editor.

Amend the code to complete the program to:

- convert the user input to integer
- check the input is between 0 and 255
- call the **getHex** subprogram, with the user input as an argument.

Do not add any further functionality.

Save your code as **Q02ciiFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 2 = 12 marks)

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6. Question 2 · part c.ii · 4 marks · Working and answer

Working and answer

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Practice workbook · Read and write text files line by line

The program opens a text file to read lines (strip, split on commas, cast, check or total) and/or writes records to a file (from input or from an array, comma-separated, one record per line), then closes the file.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-CODE-08

1. Question 5 · part c · 8 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

5 Programmers refine algorithms, use searches and read data from files.

(a) **Figure 6** shows an algorithm that determines whether the animal name inputted by the user is found.

```
1  SET a1 TO 'alpaca'
2  SET a2 TO 'bear'
3  SET a3 TO 'camel'
4  SET a4 TO 'deer'
5
6  SEND ('Enter an animal: ') TO DISPLAY
7  RECEIVE target FROM (STRING) KEYBOARD
8
9  IF (a1 = target) THEN
10     SEND ('Found') TO DISPLAY
11 ELSE
12     IF (a2 = target) THEN
13         SEND ('Found') TO DISPLAY
14     ELSE
15         IF (a3 = target) THEN
16             SEND ('Found') TO DISPLAY
17         ELSE
18             IF (a4 = target) THEN
19                 SEND ('Found') TO DISPLAY
20             END IF
21         END IF
22     END IF
23 END IF
```

Figure 6

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1. Question 5 · part c · 8 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

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Explain **one** improvement to this algorithm that will reduce the number of variables required and will enable it to work with any number of animals.

(2)

(b) Binary search is a divide and conquer algorithm.

Here is a list of data.

10	12	13	14	15	16	17	18	19	20

A binary search is used to check whether the number 11 is in the list.

The midpoint is calculated by adding the high index to the low index and **integer dividing** by 2

Complete the table to show the values for the high index, the low index and the midpoint on each pass of a binary search.

You may not need to use all the rows in the table.

(4)

high index	low index	midpoint

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1. Question 5 · part c · 8 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

(c) **Figure 7** shows the **Sales.txt** file. It stores sales information.

```
264,140,120,284,192
420,377,435,376,392
619,589,606,586,600
799,811,788,814,788
982,1007,1013,989,1009
```

Figure 7

A program is required to calculate and display:

- a subtotal for each line of sales
- a grand total for all the sales in the file.

Figure 8 shows the intended output from the program.

```
1000
2000
3000
4000
5000
Grand total: 15000
```

Figure 8

Open **Q05c** in the code editor.

Amend the code to produce the intended output.

You must use the structure and variables given in **Q05c** to complete the program.

Do not add any further functionality.

Save your code as **Q05cFINISHED** with the correct file extension for the programming language.

(8)

(Total for Question 5 = 14 marks)

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1. Question 5 · part c · 8 marks · Working and answer

Working and answer

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2. Question 3 · part a · 6 marks

2019 JUN · Paper 02

Further practice · Attempt part a only.

3 Different algorithms can be used to manipulate data.

(a) **Email.txt** contains a list of email addresses.

Open **Q03a** in the code editor.

Write a program to implement these requirements.

The code must:

- check each email address to ensure it contains the '@' symbol.
- write email addresses that do not contain the '@' symbol to an **Error.txt** file.

You must use the structure given in the file **Q03a** to complete the program.

Do not add further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(6)

(b) Explain **one** drawback of using the merge sort algorithm to sort large data sets.

(2)

(c) Open **Q03c** in the code editor.

Write a program to display the square and cube of a number between 1 and 50 entered by a user.

The code must:

- ask the user to enter a number between 1 and 50 inclusive
- display the number, the square of the number and the cube of the number, with appropriate labels
- stop when a number outside the range 1 to 50 is entered.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 3 = 14 marks)

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2. Question 3 · part a · 6 marks · Working and answer

Working and answer

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3. Question 3 · part a · 8 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

3 Several encryption algorithms have been developed.

(a) Ahmed has started to create a Caesar cipher encryption program.

Caesar cipher encryption works by giving a number value to a key.

Each letter in a plaintext message is replaced by a new ciphertext letter using the key.

For example, as shown in the table, a key value of +2 would change the plaintext message **encrypt** to the ciphertext message **gpetarv**.

Plaintext	e	n	c	r	y	p	t
Ciphertext	g	p	e	t	a	r	v

Ahmed's program already converts from plaintext to ciphertext.

He wants to extend the program to:

- accept the input of a plaintext message
- accept the input of the key, which must be a positive integer between 1 and 25
- write the generated ciphertext to a text file named **Cipher.txt**
- display the ciphertext.

Open **Q03a** in the code editor.

Complete the program code.

You must use the structure given in the file **Q03a** to complete the program.

Do not add any further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(8)

(b) Explain **one** reason why the Caesar cipher is a weak encryption algorithm.

(2)

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3. Question 3 · part a · 8 marks · Working and answer

Working and answer

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4. Question 5 · part a · 9 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

5 Julia runs a computer gaming club.

(a) She wants a program to check passwords stored in a file.

The file **passwords.txt** contains the list of passwords.

The program must:

- check each password to ensure that:
  - the first character is an uppercase letter
  - if it is, that it also includes at least one digit (0–9)
- if a password does not meet these requirements:
  - display the password
  - increment the number of incorrect passwords
- display the total number of incorrect passwords after all the passwords have been checked.

Open **Q05a** in the code editor.

Write the program.

You must use the structure given in **Q05a** to write the program.

Do not add any further functionality.

Save your code as **Q05aFINISHED** with the correct file extension for the programming language.

(9)

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4. Question 5 · part a · 9 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

(b) **Figure 3** shows an array that stores player scores after a game.

3	2	10	8	1	9

Figure 3

(i) Julia uses a bubble sort algorithm to sort the scores.

Complete the table to show how the bubble sort algorithm will sort the scores.

You may not need to use all the rows.

(3)

3	2	10	8	1	9

(ii) Explain **one** reason why a bubble sort is very efficient in terms of memory usage.

(2)

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4. Question 5 · part a · 9 marks · Working and answer

Working and answer

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5. Question 5 · part c · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

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5 A marine scientist is conducting a study of tuna in the world's oceans.

(a) A list of tuna species needs to be sorted into **descending** alphabetical order using a merge sort algorithm.

Figure 5 shows the lists created at the end of the splitting process.

Each list is a single tuna species.

Bigeye	Blackfin	Albacore	Longtail	Bluefin

Figure 5

It will require three passes to merge the lists into a single sorted list in **descending** order.

Complete the merge sort using the space provided.

(2)

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5. Question 5 · part c · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

(b) A bubble sort could be used to sort the list of tuna species into **ascending** order.

Figure 6 shows an algorithm for a bubble sort.

```
1  SET myTuna TO ["Bigeye", "Blackfin", "Albacore",
2                 "Longtail", "Bluefin"]
3  SET tmp TO 0
4  SET swaps TO True
5  SET length TO LENGTH (myTuna)
6
7  WHILE (swaps = True) DO
8      SET swaps TO False
9      FOR ndx FROM 0 TO length - 1 DO
10         IF myTuna[ndx] > myTuna[ndx + 1] THEN
11             SET tmp TO myTuna[ndx + 1]
12             SET myTuna[ndx] TO myTuna[ndx + 1]
13             SET myTuna[ndx + 1] TO tmp
14             SET swaps TO True
15         END IF
16     END FOR
17 END WHILE
```

Figure 6

There is an error in the loop between line 9 and line 16.

Complete the table to give the line number with an error and a corrected line of pseudocode.

(2)

Line number with error	
Corrected line of pseudocode	

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 2 9 3 8 A 0 1 2 1 6

5. Question 5 · part c · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

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(c) The scientist is collecting and storing data about tuna.

The data collected is:

- species
- length in centimetres
- weight in kilograms
- age in years.

The data is stored in an array.

The collected data needs to be written to a file named **TunaData.txt**

Each record stored in the file must have a code number in the first field.

Code numbers must start at 101

Each field in a record should be separated by a comma.

Figure 7 shows the contents of the file.

```
101,Yellowfin,105,15,3
102,Albacore,90,15,5
103,Skipjack,50,3,4
104,Bigeye,105,25,4
105,Atlantic Bonito,50,4,2
106,Northern Bluefin,190,120,11
107,Southern Bluefin,190,120,11
108,Tongol,90,20,4
```

Figure 7

Open **Q05c** in the code editor.

Write the program.

You must use the structure given in **Q05c** to write the program.

Do not add any further functionality.

Save your code as **Q05cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 5 = 10 marks)

5. Question 5 · part c · 6 marks · Working and answer

Working and answer

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6. Question 5 · part b · 8 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

5 Martha is a computer science student and is learning to code.

(a) **Figure 7** shows an algorithm in pseudocode written by Martha.

```
1  SET first TO 0
2  SET second TO 0
3  SET answer TO 0
4  SET user_answer TO 0
5  SET score TO 0
6
7  REPEAT 10 TIMES
8      SET score TO 0
9      SET first TO RANDOM(99)
10     SET second TO RANDOM(99)
11     SET answer TO first + second
12
13     SEND first "+" second "=" TO DISPLAY
14     RECEIVE user_answer FROM (INTEGER) KEYBOARD
15
16     IF user_answer = answer THEN
17        SET score TO score + 1
18     END IF
19 END REPEAT
20 SEND score TO DISPLAY
```

Figure 7

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6. Question 5 · part b · 8 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

There is a logic error on line 8 of the pseudocode.

(i) Explain the problem this error will cause.

(2)

(ii) Describe the purpose of the algorithm shown in **Figure 7**, assuming the error has been corrected.

(2)

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6. Question 5 · part b · 8 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

(b) Martha is writing a program to calculate grades.

Each student takes three tests. Each test has a maximum score of 50.

Scores are stored in a two-dimensional array called tbl_scores.

The lowest of the three scores is ignored and the other two are added to get a total out of 100.

Grades are based on the table in **Figure 8**.

Total	Grade
80 or more	Distinction
65 to 79	Merit
50 to 64	Pass
49 or less	Fail

Figure 8

Each student's total and grade are saved to a text file.

For example, the first student scores 45, 42 and 43. The lowest score 42 is ignored. The total is 88 and the grade is Distinction.

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6. Question 5 · part b · 8 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

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Figure 9 shows part of the text file.

```
88  Distinction
77  Merit
96  Distinction
67  Merit
48  Fail
76  Merit
```

Figure 9

The program must:

- calculate the total and grade for each student
- write total and grade to a new line in the file **grades.txt**
- display a message to say the grades file has been created.

Open **Q05b** in the code editor.

Write the program.

You must use the array given in **Q05b**.

Do not add any further functionality.

Save your code as **Q05bFINISHED** with the correct file extension for the programming language.

(8)

(Total for Question 5 = 12 marks)

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6. Question 5 · part b · 8 marks · Working and answer

Working and answer

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Practice workbook · The final 20-mark program

The last question of Paper 2: a multi-requirement program, usually built on 1D/2D or parallel arrays of test data, sometimes with a menu, subprograms, validation, searching or ordering; 11 functionality marks plus a 9-mark levels band.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-CODE-09

1. Question 6 · part whole · 20 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part whole only.

6 An agricultural college has a herd of dairy cows.

Data collected for the cows is stored in arrays.

The data stored is:

- the name of the breed
- the ease of care rating for that breed (1 is highest and 3 is lowest)
- the number of cows in the herd of that breed
- the volume of milk per day for a cow of that breed.

The college wants to present the data to farmers and to recommend the best breed of cows for them.

The best breed has the highest care rating and the largest volume of milk per day for a cow of that breed.

Open the file **Q06** in the code editor.

Write a program to:

- calculate the daily volume of milk produced by each breed
- add this daily volume to the data structure tbl_dailyVolume
- display a message informing the user what each field holds, such as breed, rating, volume per cow, count and total volume
- display the data for each breed
- calculate and display the total volume of milk produced each day by the herd
- find the recommended breed
- display the recommended breed by name.

Figure 8 shows the output from a functional program.

```
Fields are: Breed, Rating, Volume (cow), Count, Volume (day)
Red Chittagong 1 7.5 6 45.0
Sussex 2 5.7 3 17.1
Dexter 3 11.4 8 91.2
Abondance 2 11.4 7 79.8
Sahiwal 3 22.0 6 132.0
Vorderwald 1 15.2 4 60.8
Ayrshire 2 21.0 3 63.0
Jersey 1 18.3 7 128.1
Randall 2 19.0 3 57.0
Alderney 1 9.0 3 27.0
Carora 3 23.1 4 92.4
Gloucester 2 16.0 7 112.0
Total: 905.4 litres
Recommended breed: Jersey rating: 1 volume 18.3
```

Figure 8

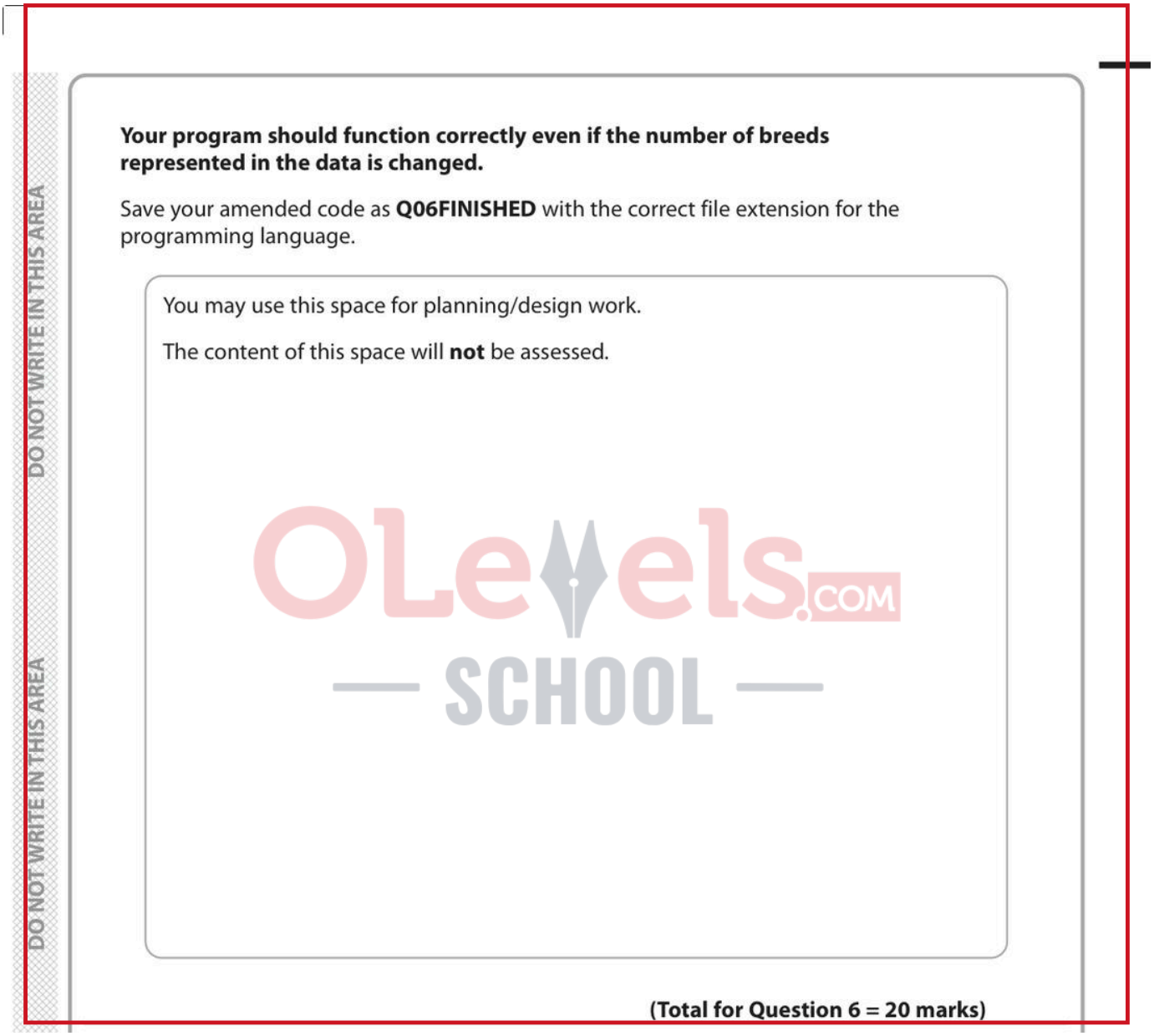
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P 7 2 9 3 8 A 0 1 4 1 6

1. Question 6 · part whole · 20 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part whole only.



Your program should function correctly even if the number of breeds represented in the data is changed.

Save your amended code as **Q06FINISHED** with the correct file extension for the programming language.

You may use this space for planning/design work.

The content of this space will **not** be assessed.

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(Total for Question 6 = 20 marks)

1. Question 6 · part whole · 20 marks · Working and answer

Working and answer

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2. Question 5 · part whole · 20 marks

2019 JUN · Paper 02

Further practice · Attempt part whole only.

5 Ria is a school librarian.

She wants a program to analyse pupil use of the library.

She wants to encourage reading by awarding gold, silver and bronze medals to the three pupils who have read the most books.

Test data has been included in the code.

Open **Q05** in the code editor.

Write a program to calculate and display:

- the total number and average number of books pupils have read
- the IDs of pupils who have read fewer than ten books
- the details of the gold, silver and bronze medal winners.

Your program should function correctly even if the number of pupils in the file is changed.

Save your code as **Q05FINISHED** with the correct file extension for the programming language.

(20)

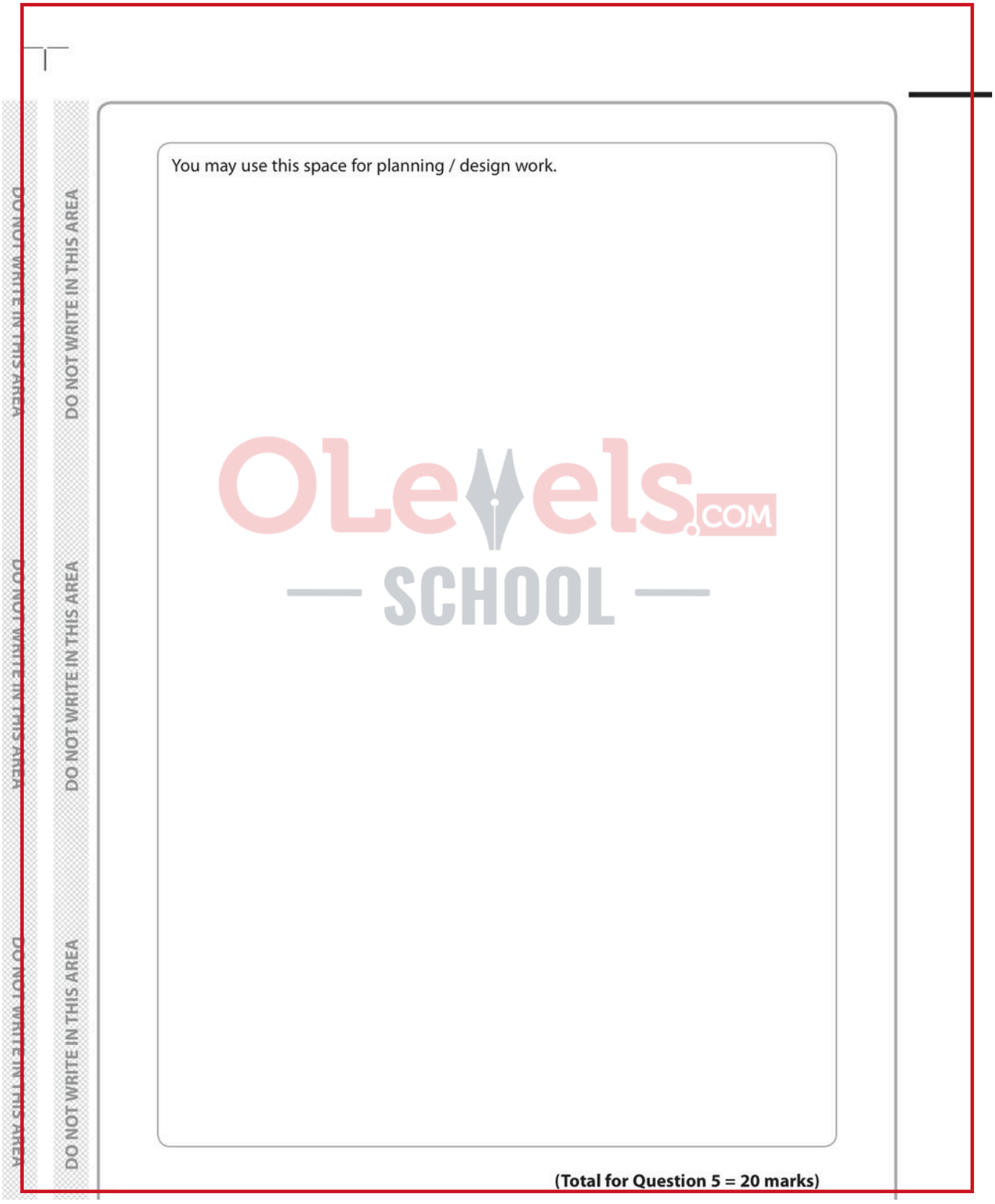
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2. Question 5 · part whole · 20 marks

2019 JUN · Paper 02

Further practice · Attempt part whole only.



You may use this space for planning / design work.

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(Total for Question 5 = 20 marks)

2. Question 5 · part whole · 20 marks · Working and answer

Working and answer

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3. Question 6 · part whole · 20 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

6 Farshia is the regional manager for an insurance company.

She manages a team of sales staff.

She wants a program to analyse the performance of her team over a number of months.

Test data has been included in the code.

Open **Q06** in the code editor.

Write a program to:

- calculate and display the total sales made by each member of the team
- calculate and display the total sales made by the whole team
- display the first name, last name and the total sales made by the two members of the team with the highest total sales. (Ignore the possibility of two or more members of the team having the same total sales.)

Your program should function correctly even if the number of months or number of members of the team is changed.

Save your code as **Q06FINISHED** with the correct file extension for the programming language.

(20)

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3. Question 6 · part whole · 20 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

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You may use this space for planning / design work.

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(Total for Question 6 = 20 marks)

3. Question 6 · part whole · 20 marks · Working and answer

Working and answer

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4. Question 5 · part whole · 20 marks

2021 JUN · Paper 02

Further practice · Attempt part whole only.

5 Bianca has started to write a program.

The program is a guessing game about countries and their capital cities.

Open **Q05** in the code editor.

The program already:

- displays this menu

```
                    Menu
----------------------------------------
[1] Add player name
[2] Play guess the capital city
[3] End game
----------------------------------------
Input your menu choice:
```

- asks for the user's menu choice.

Write the program code for the menu choices.

Menu choice [1] Add player name

If a player chooses this option, then they must input a player name.

Menu choice [2] Play guess the capital city

If a player chooses this option, then:

- they must answer five questions
- they select a question to answer by choosing a valid question number
- each question can only be selected once (they must not be able to choose a number more than once)
- the question should be displayed
- they must input the name of the capital city
- if they guess correctly their score must be incremented by 1
- if they guess incorrectly the correct answer must be displayed.

Menu choice [3] End game

If the user chooses this option, then the player name and score should be displayed.

Your program should function correctly even if the number of countries and capital cities in the file is increased.

Save your code as **Q05FINISHED** with the correct file extension for the programming language.

(20)

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4. Question 5 · part whole · 20 marks

2021 JUN · Paper 02

Further practice · Attempt part whole only.

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You may use this space for planning / design work.

The content of this page will **not** be assessed.

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(Total for Question 5 = 20 marks)

4. Question 5 · part whole · 20 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 5 · part whole · 20 marks

2021 NOV · Paper 02

Further practice · Attempt part whole only.

5 Ann likes to create programs.

Open **Q05** in the code editor.

It contains an array of words.

Ann wants the program to:

- accept the input of a word, or the number 1 to exit. No validation of the input is required
- display all the words in the array that have the same first letter as the word that has been input
- calculate and display the number of words in the array that begin with the same letter as the word that has been input
- display all the words in the array that contain the word that has been input
- calculate and display
  - the number of words in the array that contain the word that has been input
  - the number of letters in the longest word that contains the word that has been input
  - the number of letters in the shortest word that contains the word that has been input
- display
  - the longest word that contains the word that has been input
  - the shortest word that contains the word that has been input
- repeat until the user inputs the number 1 to exit the program.

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5. Question 5 · part whole · 20 marks

2021 NOV · Paper 02

Further practice · Attempt part whole only.

Figure 4 shows the display when the word **win** is input.

```
Enter a word or 1 to exit: win
----------
wind
wire

2 word(s) begin with w
----------
wind
borrowing
swing

3 word(s) contain win
The longest word has 9 letters
The shortest word has 4 letters
The longest word is borrowing
The shortest word is wind
----------
Enter a word or 1 to exit:
```

Figure 4

Write the code for the program.

Your program should function correctly even if the number of words in the array is changed.

Save your code as **Q05FINISHED** with the correct file extension for the programming language.

The next page may be used for planning / design work.

(20)

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5. Question 5 · part whole · 20 marks

2021 NOV · Paper 02

Further practice · Attempt part whole only.

You may use this space for planning / design work.

The content of this page will **not** be assessed.

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(Total for Question 5 = 20 marks)

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5. Question 5 · part whole · 20 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 6 · part whole · 20 marks

2022 JUN · Paper 02

Further practice · Attempt part whole only.

6 Carlos wants you to create a **guess the animal** game.

Open **Q06** in the code editor.

The code contains an array of animals.

It also contains a function that randomly selects an animal from the array. This is the secret word the user needs to guess.

Carlos wants the program to:

- generate the number of attempts the user has to guess the secret word.
  The maximum number of attempts is the length of the secret word +3. For example, the user has 8 attempts to guess when the secret word is **tiger**
- keep track of letters from incorrect attempts that are in the secret word and those that are not. There should be no duplicated letters
- display a message telling the user:
  - the number of letters in the secret word
  - how many attempts they have left
- force the user to input a word that is the same length as the secret word
- check whether the input word matches the secret word:
  - if the words match then a message that includes the secret word and the number of attempts taken to guess it is displayed
  - if the words do not match then:
    - letters from the attempt that appear in the secret word should be added to the correct letters store
    - letters from the attempt that do not appear in the secret word should be added to the wrong letters store
    - the contents of the correct and wrong letter stores are displayed
- allow the user another attempt until they have guessed the word or have run out of attempts
- display a message telling the user the game is over including the random word if the maximum attempts have been taken and the word has not been guessed.

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6. Question 6 · part whole · 20 marks

2022 JUN · Paper 02

Further practice · Attempt part whole only.

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Figure 4 shows the contents of the correct and wrong stores after two attempts to guess the secret word **cow**.

Secret word cow			
First attempt		**Second attempt**	
input	dog	**input**	cat
correct store	o	**correct store**	o c
wrong store	d g	**wrong store**	d g a t

Figure 4

Your program should include at least two subprograms that you have written yourself.

You must include comments in the code to explain the logic of your solution.

Save your code as **Q06FINISHED** with the correct file extension for the programming language.

You may use this space for planning/design work.

The content of this space will **not** be assessed.

(Total for Question 6 = 20 marks)

6. Question 6 · part whole · 20 marks · Working and answer

Working and answer

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7. Question 6 · part whole · 20 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

6 A program stores pairs of words in a two-dimensional array.

Each word in a pair starts with a different letter.

Each pair of words should be in alphabetical order, but some are not.

The last pair in the array is an empty pair.

The program must:

- replace the final blank pair of words with the variables **word1** and **word2**
- display the pair number of each pair, followed by each word in the pair, without punctuation
- display the longer word in the pair, indented
- display any pair found to be not in alphabetical order, in alphabetical order, indented, without punctuation.

Figure 9 shows part of the intended output from a functional program.

```
1 apple banana
      banana
2 wrist leg
      wrist
      leg wrist
3 blue yellow
      yellow
4 speaker keyboard
      keyboard
      keyboard speaker
```

Figure 9

Open the file **Q06** in the code editor.

Write a program to produce the intended output.

You must use the structure and variables given in **Q06** to complete the program.

Your program should function correctly even if the number of pairs in the array is changed.

You should use techniques to make your code easy to read.

Save your amended code as **Q06FINISHED** with the correct file extension for the programming language.

(20)

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P 7 5 7 3 7 A 0 1 5 1 6

15

Turn over ▶

7. Question 6 · part whole · 20 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

You may use this space for planning/design work.

The content of this space will **not** be assessed.

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(Total for Question 6 = 20 marks)

7. Question 6 · part whole · 20 marks · Working and answer

Working and answer

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8. Question 6 · part whole · 20 marks

2025 JUN · Paper 02

Further practice · Attempt part whole only.

6 Environmental scientists monitor the annual average temperatures of major cities around the world.

City names and their annual average temperatures are stored in two arrays.

- tbl_city stores the city names.
- tbl_temp stores the annual average temperature for each city in either Celsius or Fahrenheit. A temperature of 25° Celsius is stored as **C25** and a temperature of 80° Fahrenheit is stored as **F80**.

Scientists want a program that will enable them to:

- update temperature data already stored
- calculate the overall average annual temperature of all the cities
- view the data in a table as shown in **Figure 10**.

City	Celsius	Fahrenheit
Reggane	39	103
Cairo	28	82
New Delhi	34	93
Muscat	35	95
Athens	29	84
Barcelona	26	79
Havana	29	84
Phoenix	35	95
Brisbane	25	77
Darwin	27	81
Overall Average	30	87

Figure 10

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8. Question 6 · part whole · 20 marks

2025 JUN · Paper 02

Further practice · Attempt part whole only.

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The program must:

- allow a scientist to input the name of a city and its temperature in either Celsius or Fahrenheit
  - if the city is in the array, its temperature is updated
  - if the city is not in the array, an appropriate error message is given
- display a table with:
  - a header row and dividing line
  - each city and its annual average temperature in both Celsius and Fahrenheit shown as whole numbers
  - the overall average annual temperature for all cities in both Celsius and Fahrenheit shown as whole numbers

The formula to convert from Celsius to Fahrenheit is: F = C x 1.8 + 32

The formula to convert from Fahrenheit to Celsius is: C = (F – 32) ÷ 1.8

Open the file **Q06** in the code editor.

Write the code to produce this program with the intended output.

Include at least one subprogram.

Your program should function correctly even if the number of cities represented in the data is changed.

Use techniques to make your code easy to read.

Save your amended code as **Q06FINISHED** with the correct file extension for the programming language.

(Total for Question 6 = 20 marks)

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8. Question 6 · part whole · 20 marks · Working and answer

Working and answer

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